

- 7 (a) It is given that $y = \operatorname{sech}^{-1}\left(x + \frac{1}{2}\right)$.

Express $\cosh y$ in terms of x and hence show that $\sinh y \frac{dy}{dx} = -\frac{1}{\left(x + \frac{1}{2}\right)^2}$. [3]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

- (b) Find the first three terms in the Maclaurin's series for $\operatorname{sech}^{-1}\left(x + \frac{1}{2}\right)$ in the form

$$\ln a + bx + cx^2,$$

where a , b and c are constants to be determined. [7]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....